AQUEN FLUID TECHNOLOGIES LLP

Complete Water & Wastewater Treatment Solutions

Design | Manufacturing | Supply | Installation | Commissioning | AMC

DEGASSER TOWER

CO_2 REMOVAL SYSTEM

EFFICIENT REMOVAL OF DISSOLVED CARBON DIOXIDE (CO_2)

FRP / MSEP / MSRL VESSELS | 100 LPH – 10,000 LPH

- Reduces CO_2 and improves pH
- Prevents corrosion in downstream system
- FRP, MSEP & MSRL options
- Compact & robust design
- Low maintenance
- Suitable for industrial & boiler applications
- Custom engineered solutions



FRP DEGASSER TOWER
(100 LPH – 5,000 LPH)
with FRP Vessel & PP Internals

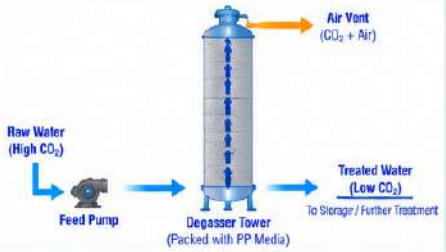


MSEP DEGASSER TOWER
(1,000 LPH – 10,000 LPH)
MS Vessel with FRP / Epoxy Lining



MSRL DEGASSER TOWER
(500 LPH – 10,000 LPH)
MS Rubber Lined Vessel

PROCESS FLOW DIAGRAM – DEGASSER TOWER SYSTEM



INTERNAL PACKING OPTIONS



PP Pall Ring



PP Structured Packing



PP Raschig Ring



PVC/PP Honeycomb
(High Efficiency)

WHY DEGASSER TOWER ?

Degasser Tower is used to remove dissolved Carbon Dioxide (CO_2) from water by air stripping process. This helps in increasing the pH of water, reducing corrosiveness and protecting downstream equipment such as RO, DM Plant, Boilers and Piping from corrosion and scaling.



KEY FEATURES

- Efficient CO_2 removal (up to 90–99%)
- FRP, MSEP and MSRL vessel options
- PP structured packing / Pall rings
- Low pressure drop & energy requirement
- Compact design with small footprint
- Suitable for industrial water, boiler feed and RO feed
- Automatic / manual operation options
- Easy installation and low maintenance
- Customised design for site conditions

TYPICAL INLET & OUTLET PARAMETERS

Parameter	Unit	Typical Inlet (Raw Water)	Typical Outlet (After Degasser)
pH	–	5.5 – 6.5	6.8 – 7.5
Total Hardness (as $CaCO_3$)	mg/L	50 – 500	50 – 500 (same)
Calcium Hardness (as $CaCO_3$)	mg/L	20 – 300	20 – 300 (same)
Dissolved CO_2	mg/L	20 – 100	< 1 – 5
Alkalinity (as $CaCO_3$)	mg/L	0 – 50	0 – 10
TDS	mg/L	100 – 2,000	100 – 2,000 (same)
Temperature	°C	15 – 40	15 – 40

APPLICATIONS / INDUSTRIES



Boiler Feed Water



RO / DM Plant



Power Plants



Pharmaceutical



Food & Beverage



Chemical Industry



Textile Industry



Steel & Metal Processing



Cooling Water Systems

TECHNICAL SPECIFICATIONS & DIMENSIONS – DEGASSER TOWER (100 LPH TO 10,000 LPH)

Model	Flow Rate (LPH)	Vessel Type (FRP / MSEP / MSRL)	Tower Diameter (mm)	Tower Height (mm)	Packing Height (mm)	Packing Type	Air Blower (m^3/hr @ m WC)	Inlet / Outlet Size (NB)
AQU-DG-0.1	100	FRP	200	1,500	1,200	PP Pall Ring	10 @ 100	15
AQU-DG-0.25	250	FRP	300	2,000	1,600	PP Pall Ring	20 @ 100	20
AQU-DG-0.5	500	FRP	400	2,500	2,000	PP Structured	40 @ 150	25
AQU-DG-1.0	1,000	FRP / MSEP	500	3,000	2,400	PP Structured	70 @ 150	32
AQU-DG-2.0	2,000	MSEP / MSRL	700	4,000	3,000	PP Structured	120 @ 200	40
AQU-DG-3.0	3,000	MSEP / MSRL	800	4,500	3,500	PP Structured	180 @ 200	50
AQU-DG-5.0	5,000	MSEP / MSRL	1,000	5,500	4,500	PP Structured	300 @ 250	65
AQU-DG-7.5	7,500	MSEP / MSRL	1,200	6,500	5,500	PP Structured	450 @ 250	80
AQU-DG-10	10,000	MSEP / MSRL	1,400	7,500	6,000	PP Structured	600 @ 300	100

Note: Dimensions are indicative and can be customized as per site conditions, inlet CO_2, temperature and desired outlet pH.

ON GOING SITE PROJECTS



FRP Degasser Tower
(500 LPH)
Installation in Progress



MSEP Degasser Tower
(2,000 LPH)
Site Installation



MSRL Degasser Tower
(10,000 LPH)
Ongoing Project

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CLEANER
WATER
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TOMORROW

1. FRP DEGASSER TOWER – 250 LPH
SINGLE VESSEL SYSTEM (COMPACT SKID MOUNTED)



FRP DEGASSER TOWER 250 LPH

DESIGN PARAMETERS	
Flow Rate	250 LPH
Vessel Type	FRP (Vertical)
Tower Diameter	300 mm
Tower Height	2,000 mm
Packing	PP Pall Ring
Air Vent	Top Mounted
MOC (Piping)	UPVC / CPVC
Control Panel	Included
Skid	MS with FRP Coating

TYPICAL INLET & OUTLET PARAMETERS		
Parameter	Inlet (Raw Water)	Outlet (After Degasser)
Dissolved CO_2	20 – 50 mg/L	< 1 – 5 mg/L
pH	5.5 – 6.5	6.8 – 7.5
Temperature	15 – 35 °C	Same

2. FRP DEGASSER TOWER – 2,000 LPH
SINGLE VESSEL SYSTEM (OUTDOOR INSTALLATION)



FRP DEGASSER TOWER 2,000 LPH

DESIGN PARAMETERS	
Flow Rate	2,000 LPH
Vessel Type	FRP (Vertical)
Tower Diameter	600 mm
Tower Height	4,000 mm
Packing	PP Structured Packing
Air Vent	Top Mounted
MOC (Piping)	UPVC / CPVC
Control Panel	Included
Platform	MS with FRP Coating

TYPICAL INLET & OUTLET PARAMETERS		
Parameter	Inlet (Raw Water)	Outlet (After Degasser)
Dissolved CO_2	20 – 100 mg/L	< 1 – 5 mg/L
pH	5.5 – 6.5	6.8 – 7.5
Temperature	15 – 40 °C	Same

3. MSEP DEGASSER TOWER – 5,000 LPH
SINGLE VESSEL SYSTEM (EPOXY LINED)



MSEP DEGASSER TOWER 5,000 LPH

DESIGN PARAMETERS	
Flow Rate	5,000 LPH
Vessel Type	MSEP (Vertical)
Tower Diameter	1,000 mm
Tower Height	5,500 mm
Internal Lining	Epoxy Lining
Packing	PP Structured Packing
Air Vent	Top Mounted
MOC (Piping)	MS/UPVC/CPVC/SS (as req.)
Control Panel	Included
Platform	MS with Safety Railing

TYPICAL INLET & OUTLET PARAMETERS		
Parameter	Inlet (Raw Water)	Outlet (After Degasser)
Dissolved CO_2	20 – 150 mg/L	< 1 – 5 mg/L
pH	5.0 – 6.5	6.8 – 7.5
Temperature	15 – 40 °C	Same

4. MSRL DEGASSER TOWER – 10,000 LPH
SINGLE VESSEL SYSTEM (RUBBER LINED)



MSRL DEGASSER TOWER 10,000 LPH

DESIGN PARAMETERS	
Flow Rate	10,000 LPH
Vessel Type	MSRL (Vertical)
Tower Diameter	1,400 mm
Tower Height	7,500 mm
Internal Lining	Rubber Lining
Packing	PP Structured Packing
Air Vent	Top Mounted
MOC (Piping)	MS/UPVC/CPVC/SS (as req.)
Control Panel	Included
Platform	MS with Safety Railing

TYPICAL INLET & OUTLET PARAMETERS		
Parameter	Inlet (Raw Water)	Outlet (After Degasser)
Dissolved CO_2	20 – 200 mg/L	< 1 – 5 mg/L
pH	5.0 – 6.5	6.8 – 7.5
Temperature	15 – 40 °C	Same